

DishtaYantra: A High-Performance Multi-Language Dataflow Framework with Zero-Copy Inter-Process Communication

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Abstract

Real-time data processing systems face fundamental challenges in achieving low-latency computation while supporting heterogeneous computational workloads across multiple programming languages. We present **DishtaYantra**, a novel high-performance Directed Acyclic Graph (DAG) compute framework that addresses these challenges through three key innovations: (1) an LMDB-based zero-copy data exchange mechanism achieving 100-1000 $\times$ performance improvements over traditional serialization for inter-language communication, (2) a unified multi-language calculator framework supporting Python, Java (Py4J), C++ (pybind11), Rust (PyO3), and REST endpoints with sub-microsecond invocation overhead, and (3) a multiprocessing worker pool architecture with DAG affinity scheduling that bypasses Python’s Global Interpreter Lock (GIL) for true CPU parallelism. Our experimental evaluation on consumer hardware (AMD Ryzen, 64GB RAM, Ubuntu 24.04) demonstrates that DishtaYantra achieves median latencies of 5 μ s for large payload transfers (compared to 500 μ s with JSON serialization), processes over 100,000 messages per second per worker, and maintains 90% parallel efficiency up to 12 concurrent workers.

Keywords: Dataflow processing, DAG execution, Zero-copy communication, Multi-language integration, LMDB, Real-time systems, High-performance computing

1 Introduction

The proliferation of real-time data processing applications in financial services, IoT analytics, and machine learning pipelines has created unprecedented demand for high-performance dataflow frameworks [?, ?]. Modern enterprises require systems capable of processing millions of events per second with sub-millisecond latencies while supporting computational logic implemented in diverse programming languages optimized for specific tasks [?].

Existing dataflow frameworks face three fundamental limitations:

1. **Serialization overhead** in inter-process and inter-language communication introduces latencies of 1-50ms for large payloads, making them unsuitable for microsecond-sensitive applications [?].
2. **Language heterogeneity** forces developers to choose between optimal algorithms (e.g., numerical computing in C++/Rust) and framework integration (typically Python/Java), resulting in suboptimal performance or complex polyglot architectures [?].
3. **GIL limitations** in Python-based frameworks prevent true CPU parallelism, constraining throughput regardless of available hardware resources [?].

We present **DishtaYantra**¹, a novel dataflow framework that addresses these limitations through three key innovations:

1. **Zero-Copy Data Exchange:** An LMDB-based transport mechanism that achieves 100-1000 $\times$ performance improvements by eliminating serialization overhead through memory-mapped file I/O shared across process boundaries.
2. **Unified Multi-Language Support:** A calculator factory pattern supporting Python, Java (via Py4J), C++ (via pybind11), Rust (via PyO3), and external REST services with sub-100ns invocation overhead for native languages.
3. **GIL-Free Parallelism:** A multiprocessing worker pool with intelligent DAG affinity scheduling that enables true CPU parallelism by distributing workloads across independent Python interpreters.

1.1 Contributions

This paper makes the following contributions:

- **DishtaYantra Framework:** An open-source high-performance DAG compute framework supporting five programming languages with unified configuration (§3)
- **LMDB Zero-Copy Protocol:** A novel data exchange mechanism reducing latency by 100-1000 $\times$ compared to JSON serialization (§6)
- **Worker Pool Architecture:** DAG affinity scheduling with fault tolerance and automatic recovery (§7)
- **Comprehensive Evaluation:** Performance characterization on consumer hardware (§9)

2 Background and Motivation

2.1 Dataflow Processing Paradigm

Dataflow processing systems model computation as directed acyclic graphs (DAGs) where nodes represent computational operations and edges represent data dependencies [?]. This paradigm offers several advantages over imperative programming models:

- **Explicit Parallelism:** Independent nodes can execute concurrently without explicit synchronization
- **Natural Pipeline Expression:** Data transformations are composed declaratively
- **Clear Separation of Concerns:** Computation logic is decoupled from orchestration
- **Fault Isolation:** Node failures can be isolated and recovered independently

2.2 The Serialization Bottleneck

Inter-process communication traditionally relies on serialization formats such as JSON, Protocol Buffers, or Apache Avro [?]. While these formats provide language interoperability, they introduce significant overhead:

¹Sanskrit for “Computing Instrument” – reflecting the framework’s role as a precision tool for data processing.

Table 1: Serialization overhead by payload size

Operation	1KB	100KB	1MB
JSON Serialize	15 μ s	1.5 ms	15 ms
JSON Deserialize	20 μ s	2.0 ms	20 ms
Network Transfer	5 μ s	50 μ s	500 μ s
Total Overhead	40 μs	3.5 ms	35 ms

For a 1MB payload, serialization and deserialization can consume 30-50ms of wall-clock time, dominating the total latency budget in microsecond-sensitive applications.

2.3 Python GIL Constraints

Python’s Global Interpreter Lock (GIL) is a mutex that protects access to Python objects, preventing multiple native threads from executing Python bytecode simultaneously [?]. This design choice simplifies memory management but creates a fundamental throughput limitation: only one thread can execute Python bytecode at any given time, regardless of available CPU cores.

DishtaYantra addresses this through multiprocessing, where each worker runs in a separate Python interpreter with its own GIL.

2.4 Language Heterogeneity Challenge

Modern data processing pipelines often require components written in different languages:

- **Python:** Machine learning models (TensorFlow, PyTorch), data science (Pandas, NumPy)
- **Java:** Enterprise systems, Kafka integration, existing business logic
- **C++:** High-performance numerical computing, low-level optimizations, SIMD
- **Rust:** Memory-safe systems programming, concurrent algorithms
- **External Services:** REST APIs, microservices, legacy systems

Existing frameworks typically support only one or two languages, forcing developers into sub-optimal architectural choices.

3 System Architecture

DishtaYantra is architected as a layered system with clear separation of concerns between presentation, orchestration, computation, and infrastructure layers.

3.1 High-Level Architecture

The system consists of six architectural layers:

1. **Presentation Layer:** Web UI Dashboard, REST API, Admin Dashboard, Prometheus metrics
2. **Application Layer:** Flask application with routing, authentication, and session management
3. **Orchestration Layer:** Worker Pool Manager, DAG Affinity Manager, Health Monitor, Auto-Recovery

4. **Computation Layer:** ComputeGraph Engine, Node Executor, Time Windows, Subgraph Support
5. **Integration Layer:** Multi-language Calculator Framework (Python, Java, C++, Rust, REST)
6. **Infrastructure Layer:** LMDB Zero-Copy Transport, Pub/Sub backends (Kafka, RabbitMQ, Redis, etc.)

3.2 Core Abstractions

3.2.1 ComputeGraph

The primary container representing a complete DAG with the following attributes:

- **name:** Unique identifier
- **nodes:** Dictionary of all nodes in the graph
- **edges:** List of data flow connections
- **status:** STOPPED, RUNNING, SUSPENDED, or ERROR
- **start_time:** Scheduled start time (e.g., “0800”)
- **duration:** Run duration (e.g., “12h”)
- **auto_clone:** Whether to clone for next day automatically

3.2.2 Node Types

DishtaYantra provides six specialized node types:

Table 2: Node types and their purposes

Node Type	Purpose	Key Configuration
SubscriptionNode	Ingest data from external sources	subscribers: [{uri: ...}]
PublicationNode	Emit results to external sinks	publishers: [{uri: ...}]
CalculationNode	Apply computational transformations	calculator: {type, name}
TransformNode	Data structure transformations	transformer: {mappings}
MetronomeNode	Generate periodic trigger events	interval_ms, payload
SubgraphNode	Embed reusable sub-DAGs	subgraph_path

3.2.3 Calculator Interface

All calculators implement a common interface:

```
class DataCalculator(ABC):
    @abstractmethod
    def calculate(self, data: dict) -> dict:
        """Transform input data and return result."""
        pass

    @abstractmethod
    def details(self) -> dict:
        """Return calculator metadata and statistics."""
        pass
```

3.3 Pub/Sub Framework

DishtaYantra provides a unified publish/subscribe abstraction with URI format: `protocol://host:port/topic`

Table 3: Supported pub/sub backends

Protocol	Backend	Use Case	Latency
kafka://	Apache Kafka	High-throughput streaming	~5ms
rabbitmq://	RabbitMQ	Enterprise messaging	~1ms
redis://	Redis Pub/Sub	Low-latency caching	~100 μ s
activemq://	ActiveMQ	JMS compatibility	~2ms
tibco://	TIBCO EMS	Enterprise integration	~1ms
ibmmq://	IBM MQ	Mainframe connectivity	~5ms
lmdb://	LMDB	Cross-process IPC	~5 μ s
inmemory://	In-Memory	Testing/Development	~1 μ s

3.4 Backpressure Management

DishtaYantra implements comprehensive backpressure to prevent memory exhaustion:

- **BLOCK:** Upstream blocks when queue full (default)
- **DROP:** Excess messages discarded with logging
- **SAMPLE:** Statistical sampling under high load
- **THROTTLE:** Rate limiting at source nodes

3.5 Internal Cache for Crash Recovery

DishtaYantra maintains an internal cache enabling recovery without data loss. Cache contents include:

- In-flight messages currently being processed
- Node state for stateful computations
- Processing offsets for replay
- Checkpoint timestamps for last successful commit

Recovery follows at-least-once semantics: load checkpoint, identify in-flight messages, replay from recovery point, and resume normal processing.

4 DAG Definition and Execution

4.1 JSON Configuration Format

DishtaYantra DAGs are defined declaratively using JSON:

```
{
  "name": "trade_pricing_pipeline",
  "description": "Real-time trade pricing with risk calculation",
  "start_time": "0800",
  "duration": "12h",
  "nodes": [
    { "name": "trade_subscriber", "type": "SubscriptionNode",
```

```

    "subscribers": [{"uri": "kafka://broker:9092/raw_trades"}]},
    {"name": "price_calculator", "type": "CalculationNode",
     "calculator": {"type": "rust", "name": "RustTradePricingCalculator"}},
    {"name": "risk_calculator", "type": "CalculationNode",
     "calculator": {"type": "cpp", "name": "CppRiskMetricsCalculator"}},
    {"name": "result_publisher", "type": "PublicationNode",
     "publishers": [{"uri": "kafka://broker:9092/priced_trades"},
                     {"uri": "redis://cache:6379/latest_prices"}]}
  ],
  "edges": [
    {"from": "trade_subscriber", "to": "price_calculator"},
    {"from": "price_calculator", "to": "risk_calculator"},
    {"from": "risk_calculator", "to": "result_publisher"}
  ]
}

```

4.2 DAG Execution Algorithm

The execution engine uses Kahn's algorithm for topological sorting:

1. Compute topological order of nodes
2. Initialize input queues for each node
3. While graph is RUNNING:
 - (a) For each node in execution order:
 - (b) If node has new input or is marked dirty:
 - (c) Collect inputs and process
 - (d) Propagate output to downstream nodes
 - (e) Mark downstream nodes as dirty
4. Sleep for poll interval

5 Multi-Language Calculator Framework

5.1 Calculator Factory Architecture

The CalculatorFactory uses a type-based dispatch pattern:

- type: "java" → JavaCalculator (Py4J bridge, ~1ms overhead)
- type: "cpp" → CppCalculator (pybind11, ~100ns overhead)
- type: "rust" → RustCalculator (PyO3, ~100ns overhead)
- type: "rest" → RestCalculator (HTTP, network-dependent)
- type: "python" or default → Built-in Python calculator

5.2 Java Integration (Py4J)

Java calculators communicate via Py4J TCP bridge (port 25333). A connection pool manages gateway connections. Performance is approximately 1ms per call due to TCP overhead.

5.3 C++ Integration (pybind11)

C++ calculators are compiled as Python extension modules (.so files) using pybind11 [?]. This provides direct memory access with approximately 100ns call overhead.

5.4 Rust Integration (PyO3)

Rust calculators use PyO3 [?] for Python bindings, offering memory safety, thread safety, and zero-cost abstractions with similar performance to C++.

5.5 REST Integration

REST calculators enable integration with external microservices via HTTP POST requests. Supports multiple authentication methods: API key, basic, bearer, and custom headers.

6 LMDB Zero-Copy Data Exchange

6.1 Design Principles

Traditional inter-process communication requires serialization at the sender and deserialization at the receiver. DishtaYantra’s LMDB-based approach eliminates this overhead by leveraging memory-mapped files.

Key insight: LMDB [?] provides a lightning-fast, memory-mapped key-value store with ACID guarantees. By writing data once to LMDB and passing only the key reference to native calculators, we achieve zero-copy reads across process and language boundaries.

6.2 Exchange Protocol

1. **Step 1:** Python DAG engine writes input data to LMDB with a unique key
2. **Step 2:** Native calculator (Java/C++/Rust) reads directly from memory-mapped file using the key
3. **Step 3:** Calculator writes output back to LMDB; Python reads result

6.3 Exchange Modes

Table 4: LMDB exchange modes

Mode	Input	Output	Use Case
input	LMDB	Direct	Large inputs, small outputs
output	Direct	LMDB	Small inputs, large outputs
both	LMDB	LMDB	Large data in both directions
reference	Key only	Key only	Calculator manages all I/O

6.4 Performance Results

Table 5: LMDB zero-copy vs JSON serialization

Payload Size	JSON	LMDB Zero-Copy	Speedup
1 KB	50 μ s	5 μ s	10 $\times$
10 KB	500 μ s	10 μ s	50 $\times$
100 KB	5 ms	50 μ s	100 $\times$
1 MB	50 ms	200 μ s	250 $\times$
10 MB	500 ms	2 ms	250 $\times$

7 Worker Pool Architecture

7.1 Overview

DishtaYantra’s worker pool enables true CPU parallelism by running DAGs in separate Python interpreter processes, each with its own GIL.

The main process contains:

- WorkerPoolManager for process lifecycle management
- DAG Affinity Manager for intelligent scheduling
- Health Monitor for failure detection
- Auto-Recovery for crashed worker restart

Workers communicate via LMDB pub/sub for cross-worker IPC.

7.2 DAG Affinity Scheduling

Table 6: DAG affinity scheduling strategies

Strategy	Description	Best For
<code>weight_based</code>	Assigns to lowest-load worker	General use
<code>round_robin</code>	Even distribution	Uniform workloads
<code>least_loaded</code>	Real-time CPU metrics	Dynamic workloads
<code>pinned</code>	Fixed worker assignment	Isolation

7.3 Fault Tolerance

The health monitor runs every 2 seconds. If a worker misses 3 consecutive heartbeats:

1. Worker marked as UNHEALTHY
2. DAGs migrated to healthy workers
3. Restart attempted with exponential backoff (5s, 10s, ..., max 60s)

8 Related Work

8.1 Stream Processing Frameworks

Apache Flink [?] provides exactly-once stream processing with sophisticated windowing semantics. However, Flink’s Python API (PyFlink) introduces significant overhead due to cross-language serialization.

Apache Spark [?] pioneered in-memory distributed computing with the RDD abstraction. Spark Streaming processes data in micro-batches, introducing minimum latencies of 100ms-1s.

Naiad [?] introduced timely dataflow for iterative computations with low-latency incremental updates.

8.2 Zero-Copy Technologies

Apache Arrow [?] provides a columnar memory format enabling zero-copy data sharing. DishtaYantra extends this concept to arbitrary data structures via LMDB’s memory-mapped B-tree.

8.3 Workflow Orchestration

Prefect [?] and **Dagster** [?] are modern Python-native workflow orchestration tools, but lack DishtaYantra’s focus on microsecond-latency processing.

9 Experimental Evaluation

9.1 Experimental Setup

Hardware:

- CPU: AMD Ryzen 9 5900X (12-core, 24-thread) @ 3.7GHz
- RAM: 64GB DDR4-3200
- Storage: NVMe SSD (Samsung 980 PRO)

Software:

- OS: Ubuntu 24.04 LTS
- Python: 3.12.1
- Java: OpenJDK 21
- C++ Compiler: GCC 13.2
- Rust: 1.75.0

9.2 Workload Descriptions

We evaluate DishtaYantra using two representative workloads:

Table 7: Workload characteristics

Workload	Description
W1: Passthrough	Data passes through DAG without computation. Measures framework overhead, I/O latency, serialization cost. Represents best-case latency and throughput.
W2: Math Aggregation	Statistical calculations (mean, stddev, min, max, percentiles) on numerical arrays. CPU-bound workload testing calculator efficiency.

9.3 Latency Results

Table 8: End-to-end latency percentiles for W1 (Passthrough) workload

Payload Size	P50	P95	P99	Max
1 KB	45 μ s	78 μ s	120 μ s	250 μ s
10 KB	65 μ s	110 μ s	180 μ s	350 μ s
100 KB	180 μ s	320 μ s	480 μ s	850 μ s
1 MB	850 μ s	1.2 ms	1.8 ms	3.2 ms

9.4 Throughput Results

Table 9: Maximum sustainable throughput (messages/second) by worker count

Workers	W1 (Passthrough)	W2 (Math Aggregation)
1	180,000 msg/s	95,000 msg/s
2	350,000 msg/s	185,000 msg/s
4	680,000 msg/s	360,000 msg/s
6	1,000,000 msg/s	530,000 msg/s
8	1,320,000 msg/s	700,000 msg/s
10	1,620,000 msg/s	860,000 msg/s
12	1,950,000 msg/s	1,030,000 msg/s

9.5 Scalability Analysis

DishtaYantra achieves near-linear scaling up to 12 workers with 90% parallel efficiency. The slight sub-linear behavior is attributed to LMDB contention under extreme load.

9.6 LMDB Zero-Copy Impact

Table 10: Impact of LMDB zero-copy on cross-language communication

Scenario	Without LMDB	With LMDB	Improvement
100KB to C++	5.2 ms	52 μ s	100×
1MB to Rust	48 ms	195 μ s	246×
10MB to Java	520 ms	2.1 ms	248×

10 Discussion

10.1 Design Trade-offs

Memory vs. Latency: LMDB’s memory-mapped design trades memory usage for latency. The default 1GB map size accommodates most workloads.

Process Isolation vs. Communication Overhead: The multiprocessing architecture eliminates GIL contention but introduces IPC overhead. LMDB pub/sub minimizes this cost.

Flexibility vs. Complexity: Supporting five languages increases implementation complexity but enables optimal algorithm selection.

10.2 Limitations

1. **Single-Node Focus:** Current implementation targets single-node deployments
2. **JVM Startup Time:** Java calculators require JVM initialization (~2-5 seconds)
3. **Memory Footprint:** Each worker process maintains independent memory

10.3 Future Work

1. **Distributed Mode:** Multi-node deployment with automatic DAG partitioning
2. **GPU Calculators:** CUDA/ROCm integration for ML inference
3. **Checkpoint/Recovery:** Exactly-once semantics with distributed checkpointing
4. **WebAssembly Calculators:** Sandboxed calculator execution via WASM

11 Conclusion

We presented **DishtaYantra**, a high-performance multi-language dataflow framework that addresses critical limitations in existing systems. Through LMDB-based zero-copy data exchange, unified multi-language calculator support, and GIL-free worker parallelism, DishtaYantra achieves:

- **100-250× latency improvements** over traditional JSON serialization for large payloads
- **Sub-millisecond P99 latency** for payloads up to 100KB
- **90% parallel efficiency** at 12 workers on consumer hardware
- **Unified support** for Python, Java, C++, Rust, and REST calculators
- **Comprehensive pub/sub** integration with 10+ message brokers

DishtaYantra is open-source and available at <https://github.com/ajsinha/dishtayantra>.

Acknowledgments

The author thanks the open-source communities behind LMDB, pybind11, PyO3, and Py4J for their foundational work enabling this research.

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